Bayes Update Cards Student Handout

TOK Cognitive Science Activity Bank • Mathematics • 35-45 min

Category	Mathematics	**Time**	35-45 min
Difficulty	Medium	**ID**	bayes-update-cards

Big idea: New evidence updates prior probability; it does not replace the background rate.

Student-Facing Prompt

Inspect Bayes Update Cards Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Bayes Update Cards Stimulus A	Student-facing stimulus 1: Fictional case: 10 of 1000 people have Condition X; a positive test catches 9 real cases and falsely flags 99 non-cases. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Bayes Update Cards Stimulus B	Student-facing stimulus 2: Notation contrast card: P(H | +) is not the same as P(+ | H). Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Bayes Update Cards Reveal / Twist	Student-facing stimulus 3: Notation contrast card: P(H | +) is not the same as P(+ | H). Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — New evidence updates prior probability; it does not replace the background rate.

Actual Lesson Questions

#	Question for students
1	After inspecting Bayes Update Cards Stimulus A, what is your first knowledge claim?
2	Which exact detail from Bayes Update Cards Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	How does notation shape mathematical knowledge?

Ready-to-Use Examples

• Fictional case: 10 of 1000 people have Condition X; a positive test catches 9 real cases and falsely flags 99 non-cases.

• Notation contrast card: P(H | +) is not the same as P(+ | H).

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Activity Steps

1. Give students a fictional condition, base rate, and test accuracy in words.
2. Ask for an intuitive estimate after a positive result.
3. Build the case using natural frequencies, separating true positives from false positives.
4. Introduce the update calculation $P(H \mid +) = 9 / (9 + 99) = 1 / 12 \approx 8.3\%$.
5. Compare how the same information feels in words, percentages, frequencies, and notation.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: How does mathematical notation change what is visible in a knowledge claim?
- Why do many knowers confuse $P(H \mid +)$ with $P(+ \mid H)$?
- What responsibilities come with communicating risk to non-specialists?

Knowledge Questions

- How does notation shape mathematical knowledge?
- When is a mathematically correct risk claim still misleading?
- What makes a representation responsible?

Transfer Example

For example, run this activity with a simple classroom version using frequency cards for a fictional screening test.. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...